

POE2 FORGE

PROJECT ROADMAP

A PoB2-replacement build optimizer with theorycrafting and market intelligence

VERSION	1.0 — Roadmap draft
AUTHOR	POE2 Forge project
CURRENT BUILD	POE2Forge_v5.html (working)
NEXT MILESTONE	v6 — Phase 0 (Shell + Welcome + shared build context)
TARGET LEAGUE	Runes of Aldur (0.5) — 2026-05-29 launch

Executive Summary

BLUF: POE2 Forge is a Path of Exile 2 build tool that combines a Path of Building 2 replacement, a live trade-market optimizer, an AI-assisted theorycrafting partner, and a market-intelligence layer in one cohesive web app. Today it functions as a basic trade-market optimizer with hardcoded character data. The 11-phase roadmap below takes it from where it is now to a full PoB2 alternative with market forecasting by mid-to-late 2026.

Vision

Replace PoB2 entirely for daily use, with the same mathematical rigor for damage and EHP calculations but a substantially cleaner UI — comparable in polish to Maxroll or Mobalytics, without the ad clutter. On top of that, two capabilities no other tool offers: an AI partner (this Claude instance) that theorycrafts cross-class S-rank build patterns in real time, and a market-intelligence layer that forecasts which items will spike or crash based on emerging build meta.

Strategic Differentiators

- ◆ **Theorycraft-to-optimize bridge.** Paste a creator's PoB2 code, immediately see what live trade-market gear would upgrade it. No competitor does this.
- ◆ **AI cross-class scanner.** Pattern-match builds for downside-text negation by ascendancy keystones — the technique behind every recent S-tier build (Lich + Last Lament, Hollow Form + Rolling Slam, etc.).
- ◆ **Live market data with historical depth.** The proxy starts logging trade prices in Phase 1, building a time-series dataset that competitors lack.
- ◆ **Prediction-accuracy feedback loop.** Theorycrafted builds get tracked over time against community findings, so the AI learns which patterns to trust.

Current State

v5 ships a three-panel HTML app with paperdoll UI, live trade API integration via a Node.js proxy on localhost:3001, an upgrade-scoring engine, and priority sliders. Character data is currently hardcoded (placeholder Druid Oracle). Phase 0 removes that placeholder and establishes the shared build-context architecture that all subsequent phases depend on.

Phase Overview

Phase	Name	Status
0	Shell + Welcome + shared build context	Next
1	POESESSIONID auth + character import + price logging	Planned
2	PoB2 decoder + Send to Optimizer	Planned

3	Build template gallery	Planned
4	Cross-class S-rank scanner	Planned
5	Advanced Thaumaturgy alt-quality reference	Planned (5/29)
6	Build export (PoB2 + in-game tracker codes)	Planned
7	Live build editor (gear / skills / tree)	Long-range
8	Damage calculation engine	Long-range
9	Build comparison & A/B testing	Long-range
10	Market intelligence & flip dashboard	Long-range
11	Build telemetry & feedback loop	Long-range

PHASE 0

Shell + Welcome + Shared Build Context

● Next up — work begins after roadmap sign-off

GOAL

Establish the architectural foundation that every later phase depends on. Remove hardcoded character data and replace it with a shared build object that both Optimizer and Theorycraft tabs read from.

SCOPE

- Create POE2Forge_v6.html from v5 as a clean rollback point
- Three-tab top-level header: WELCOME / OPTIMIZER / THEORYCRAFT
- Welcome tab with first-time setup checklist (Node.js, proxy, POESESSIONID, account)
- Shared currentBuild JavaScript state object modeled on PoB2's schema
- Optimizer tab rewired to read from currentBuild — empty state when null
- Theorycraft tab with four sub-tab placeholders for Phases 2-5
- localStorage flag for first-time vs returning user routing
- Live proxy health-check integrated into Welcome's checklist

KEY DELIVERABLES

- ◆ POE2Forge_v6.html ships with three working tabs
- ◆ Hardcoded Druid Oracle data removed; empty state displayed when no build loaded
- ◆ currentBuild schema documented and locked in for Phase 7+ compatibility

DEPENDENCIES & RISK

Dependencies	None — work begins immediately
Risk	Low. Architectural change but well-scoped.

PHASE 1

POESESSIONID Auth + Character Import + Price Logging

Planned — begins after Phase 0

GOAL

Connect Forge to the user's actual GGG account via POESESSIONID. Provide League and Character dropdowns on the Welcome tab. Import live character data into currentBuild and populate the Optimizer.

SCOPE

- Proxy endpoint to read POESESSIONID from .env and fetch character list
- Welcome tab dropdowns: League → Character → Import button
- Import flow populates currentBuild from GGG's /character endpoint
- Optimizer's paperdoll renders the imported gear
- Logout button clears currentBuild and returns to empty state
- Proxy begins logging every trade search to a local SQLite file for Phase 10 use
- OAuth-ready auth layer abstraction (POESESSIONID now, OAuth later when GGG approves)

KEY DELIVERABLES

- ◆ User can import real character data and use the Optimizer with live gear
- ◆ Trade-market data logging starts collecting historical pricing
- ◆ Auth abstraction makes the OAuth migration painless when approval arrives

DEPENDENCIES & RISK

Dependencies	Phase 0 ships. GGG account with POESESSIONID configured in .env.
Risk	Low for POESESSIONID path. OAuth migration timeline unknown (GGG approval pending).

PHASE 2

PoB2 Decoder + Send to Optimizer

Planned — Theorycraft sub-tab #1

GOAL

Paste a Path of Building 2 export code, decode it client-side, and populate currentBuild with the full build. Enable the theorycraft-to-optimize bridge — paste a creator's build, immediately scan trade market for upgrades.

SCOPE

- Theorycraft sub-tab with single textarea for PoB2 code input
- Client-side decode pipeline: base64-urlsafe-decode → zlib inflate → XML parse
- Map decoded XML into currentBuild schema (gear, skills, tree, ascendancy, stats)
- Display decoded build summary inline (skills, key gear, ascendancy choices)
- 'Send to Optimizer' button switches tabs and runs upgrade scan on the imported build
- Pako library bundled for zlib (client-side, no proxy roundtrip needed)

KEY DELIVERABLES

- ◆ Any PoB2 code becomes a fully usable build in Forge in under three seconds
- ◆ Theorycraft-to-optimize workflow unlocked — the headline feature of Phase 2

DEPENDENCIES & RISK

Dependencies	Phase 0 (shared currentBuild context).
Risk	Medium. PoB2's XML schema has edge cases; some build types may not decode cleanly initially.

PHASE 3

Build Template Gallery

Planned — Theorycraft sub-tab #2

GOAL

Curated, searchable library of launch-ready builds for the current league. Each template is a stored PoB2 code that decodes through Phase 2's pipeline.

SCOPE

- Card-based grid layout for build templates
- Day-one curated set: Bridget Whirling Assault Martial Artist, Wollie Facebreaker Hollow Form, Spirit Walker Tame Beast minion start, Gemling Absent-amulet caster, Blood Mage Vitality Siphon
- Click card → decode → populate currentBuild → optional auto-send to Optimizer
- localStorage persistence for user-saved custom builds alongside curated ones
- Filter by class, ascendancy, playstyle, league-start vs endgame, budget tier

KEY DELIVERABLES

- ◆ Faster discovery path than Mobalytics or Maxroll — no clicking through ad-laden pages
- ◆ User builds a personal library of bookmarked PoB codes across leagues

DEPENDENCIES & RISK

Dependencies	Phase 2 (decoder).
Risk	Low. Mostly data curation and UI work.

PHASE 4

Cross-Class S-Rank Scanner

Planned — Theorycraft sub-tab #3

GOAL

The signature analytical tool. Pattern-match builds for the 'downside text negated by ascendancy keystone' pattern that has produced every S-tier build of the last several leagues.

SCOPE

- JSON-based rule pack defining downside-negation patterns
- Day-one rules: Last Lament + Lich Eternal Life, Rolling Slam + Hollow Form clones, Vaal Pact sustained-leech detector, Time of Need + CDR scaling traps
- Scanner reads currentBuild and outputs flagged interactions with confidence scores
- Two input modes: scan an existing currentBuild, OR manually pick skill + ascendancy + unique combo
- Plain-language explanation of why each flag fires (the actual game-mechanics reasoning)

KEY DELIVERABLES

- ◆ Theorycraft new build ideas in minutes instead of waiting for creator videos
- ◆ Output is reasoning-grade — explains the interaction, not just lists it

DEPENDENCIES & RISK

Dependencies	Phase 0 (currentBuild schema). Phase 2 helpful but not required.
Risk	Medium. Rule pack quality determines tool quality. Phase 11's feedback loop refines accuracy over time.

PHASE 5

Advanced Thaumaturgy Alt-Quality Reference

Planned — Theorycraft sub-tab #4 — 5/29 deliverable

GOAL

Searchable database of every skill gem's hidden Advanced Thaumaturgy alt-quality effect. Wires into Phase 4 so the S-rank scanner can flag Gemling builds where the alt-quality negates a downside.

SCOPE

- Wait for 0.5 launch (May 29 2026), then datamine alt-quality strings from PoE2DB
- Build searchable table: skill name, normal quality effect, Advanced Thaumaturgy alt-quality effect
- Cross-reference: highlight skills where alt-quality removes a downside keyword
- Integration hook: Phase 4 scanner flags 'this Gemling build's main skill has an alt-quality that...'

KEY DELIVERABLES

- ◆ First public Advanced Thaumaturgy reference outside of in-game tooltips
- ◆ S-rank scanner gains visibility into a 100+ skill alt-quality surface area

DEPENDENCIES & RISK

Dependencies	0.5 patch live. PoE2DB datamine complete (typically 24-48h after launch).
Risk	Low — purely data work once the patch is live.

PHASE 6

Build Export (PoB2 + In-Game Tracker Codes)

Planned — closes the import/export loop

GOAL

Reverse the Phase 2 decoder. Export currentBuild back to a PoB2 paste code, and generate the upcoming 0.5 in-game build tracker code format.

SCOPE

- Serialize currentBuild → PoB2 XML → zlib deflate → base64-urlsafe encode
- Validate by round-tripping: decode → re-encode → diff (should match)
- Research and implement the in-game build tracker code format (spec pending GGG release)
- Export buttons available wherever currentBuild is non-null

KEY DELIVERABLES

- ◆ Forge is no longer a one-way street — builds flow in and out cleanly
- ◆ Builds theorycrafted in Forge can be shared back to the community in standard formats

DEPENDENCIES & RISK

Dependencies	Phase 2 (decoder). 0.5 in-game tracker spec from GGG (timing TBD).
Risk	Low for PoB2 export. In-game tracker risk depends on whether GGG publishes the format spec or keeps it in

PHASE 7

Live Build Editor (Gear / Skills / Tree)

Long-range — first PoB2-replacement phase

GOAL

Stop being read-only. Let the user modify gear mods, swap support gems, and adjust passive tree allocations inside Forge, with live stat recalculation.

SCOPE

- Gear editor — click any paperdoll slot, edit mods inline, see live recalc
- Skill/support gem editor — pick main skill, add/remove/swap support gems, view DPS deltas
- Passive tree editor — start with 'paste from passive planner' import, then build interactive tree later
- Skill gem database sourced from PoE2DB datamines
- Calc engine integration (Phase 8) provides the live recalc

KEY DELIVERABLES

- ◆ First version of Forge that can theorycraft a build from scratch end-to-end
- ◆ Daily PoB2 usage starts being replaceable for simple builds

DEPENDENCIES & RISK

Dependencies	Phase 8 (calc engine) must exist for live recalc.
Risk	High. Passive tree editor is genuinely hard. May ship in sub-phases.

PHASE 8

Damage Calculation Engine

Long-range — hardest phase technically

GOAL

The math layer that turns currentBuild into DPS, EHP, defenses, and per-stat contributions. PoB2 has years of game-mechanics reverse-engineering here.

SCOPE

- Phase 8A: Use PoB2 as a headless calculation backend. Send currentBuild to a hosted PoB2 instance, receive computed stats. User never sees PoB2's UI, but its math powers Forge.
- Phase 8B (long-range): Port PoB2's Lua damage code to JavaScript. Massive but tractable — PoB2 is open source.
- Phase 8C (rejected): Build our own from scratch. Worst option for 'PoB2-replacement' framing.

KEY DELIVERABLES

- ◆ Forge displays accurate DPS / EHP / defenses for any currentBuild
- ◆ Phase 7's editor gains the live recalc it depends on

DEPENDENCIES & RISK

Dependencies	Server infrastructure for headless PoB2 (Phase 8A) or significant engineering investment (Phase 8B).
Risk	High. The technically hardest phase on the roadmap. Choice between 8A and 8B determines ship date by m

PHASE 9

Build Comparison & A/B Testing

Long-range

GOAL

Two builds side-by-side with stat diffs, gear-cost diffs, and 'what if I swap this item' simulation. Theorycrafting becomes empirical instead of intuitive.

SCOPE

- Side-by-side build view with synchronized scrolling
- Stat-by-stat diff highlighting (green/red gain/loss indicators)
- Gear-cost diff in current league currency
- Item swap simulator: drag an item from one build's slot to another, see deltas
- Build versioning — save 'before/after' snapshots of your own theorycrafting iterations

KEY DELIVERABLES

- ◆ Theorycraft suggestions stop being 'trust me' and become 'here is the math'
- ◆ Foundation for Phase 11's prediction accuracy tracking

DEPENDENCIES & RISK

Dependencies	Phase 8 (calc engine).
Risk	Medium. UI complexity is the main challenge.

PHASE 10

Market Intelligence & Flip Dashboard

Long-range — most distinctive phase on the roadmap

GOAL

Turn the trade-data logging that started in Phase 1 into actionable market intelligence. Forecast item demand based on build meta. Surface flip opportunities.

SCOPE

- Build popularity index — aggregate which gear mods, uniques, and skill gems appear in popular builds. Sources: ladder API, public characters, our own tracked builds
- Demand forecaster — per item, compute popular builds requiring it × current supply × recent price trend → 'underpriced relative to demand' flag
- Personal flip dashboard — given the user's stash and currency, suggest specific buy-low/sell-high trades
- Meta-shift detector — when a creator drops a new build video or a patch hits, flag items about to spike
- Historical price charts — Phase 1's logged data finally gets visualized

KEY DELIVERABLES

- ◆ Forge becomes a currency-making tool, not just a build tool
- ◆ Predictive market intelligence — a category competitors don't address

DEPENDENCIES & RISK

Dependencies	Phase 1 must have been logging data for at least one full league. Phase 4's S-rank scanner feeds the meta-
Risk	Medium-high. Data quality and the noise of any in-game economy are real challenges. Per project owner: tra

PHASE 11

Build Telemetry & Feedback Loop

Long-range — closes the AI improvement loop

GOAL

Compare AI theorycraft predictions to actual community findings and gameplay results. Build a record of which patterns predict well and which don't. Weight future scans accordingly.

SCOPE

- Every theorycrafted build gets a unique ID at creation
- Predictions are tagged: 'Build X will be S-tier because of pattern Y'
- Outcome tagging: after league play, mark whether the build actually worked out
- Sources for outcome data: user observation, community posts, ladder presence, creator coverage
- Pattern accuracy stats — over time, build a record of which AI-recognized patterns predict well
- Phase 4 scanner output gets weighted by historical prediction accuracy

KEY DELIVERABLES

- ◆ AI theorycraft improves measurably league-over-league
- ◆ Genuinely novel — no other build tool has a learning feedback loop

DEPENDENCIES & RISK

Dependencies	Phases 4 and 9 shipped. Multiple leagues of data collection.
Risk	Low technically. The challenge is sustained engagement to actually log outcomes.

Next Steps

Immediate Actions

- ◆ **Sign off on this roadmap.** Phases 7-11 are long-range — flag any reordering or kills before Phase 0 begins.
- ◆ **Start Phase 0.** Build v6 with the three-tab shell, Welcome tab, and shared currentBuild context.
- ◆ **Hand off to Claude Code.** A handoff prompt is provided alongside this document so phase work can move into the project folder rather than continuing through chat.
- ◆ **Begin trade-data logging in Phase 1.** Capture pre-launch market state before 0.5 ships May 29 2026.

Open Questions

- ? **GGG OAuth approval timing.** Phase 1 uses POESESSIONID as a fallback. When approval lands, swap the auth layer.
- ? **0.5 in-game build tracker format.** Spec not yet published. Phase 6's tracker export depends on it.
- ? **Phase 8 calc engine — 8A or 8B?** Headless PoB2 backend (faster ship, dependency on PoB2 maintenance) or full Lua-to-JS port (longer ship, full ownership). Decision deferred until Phase 7 is in progress.

Closing Note

This roadmap is ambitious. Phases 0-6 are tractable in months; Phases 7-11 are multi-league efforts. The strength of the plan is that each phase ships standalone value — Phase 0 alone fixes the hardcoded-data problem, Phase 1 unlocks real character imports, Phase 2 delivers the theorycraft-to-optimize bridge that competitors don't have. No phase requires waiting for the entire roadmap to complete before producing usable output.